

# ONLINE LEARNING FOR NETWORK RESOURCE ALLOCATION

## CNI SEMINAR, IISC

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November 22, 2023



# PRESENTATION ORGANIZATION

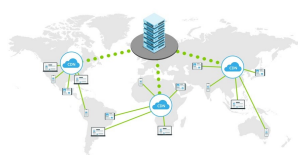
- 1 Network Resource Allocation . . . . . 3
- 2 Caching . . . . . 5
- 3 Similarity Caching . . . . . 27
- 4 Inference Delivery Networks . . . . . 44
- 5 Fairness in Dynamic Resource Allocation . . . . . 50
- 6 Concluding Remarks . . . . . 64

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# NETWORK RESOURCE ALLOCATION

Network resource allocation is ubiquitous



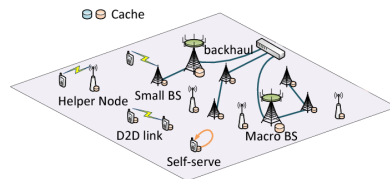
(a) CDNs



(b) ICNs



(c) Cloud computing



(d) Edge/Wireless IoT

## Goal

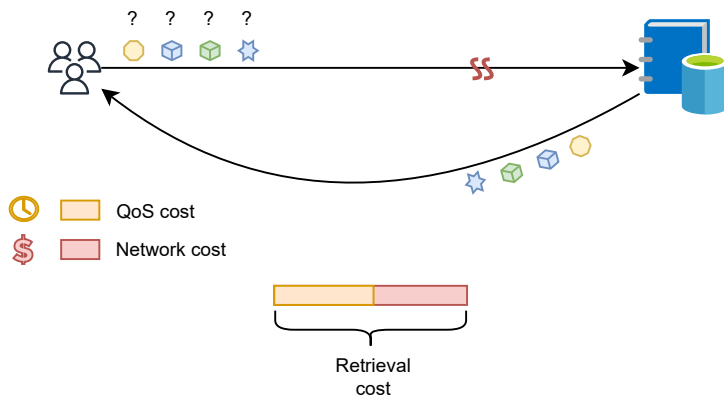
To provide *faster* service to demands generated by users (😊), or to *reduce* the computation or communication load on the system (\$).



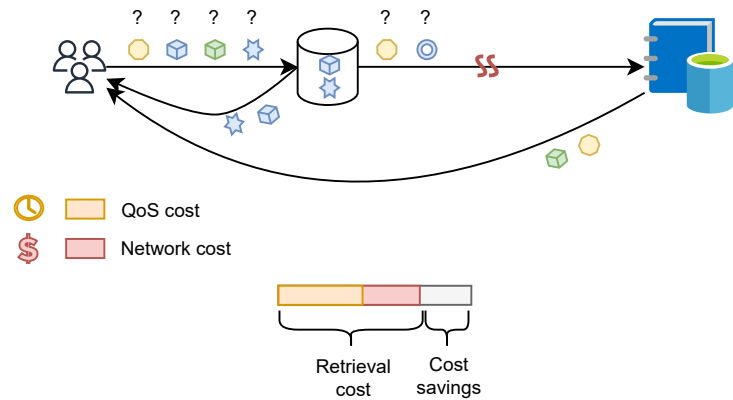
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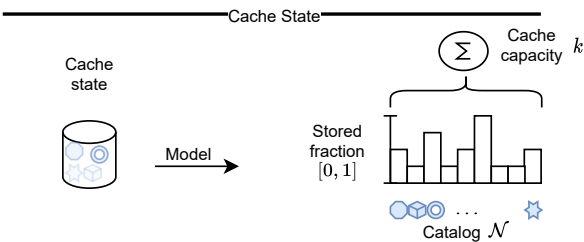
# CACHING



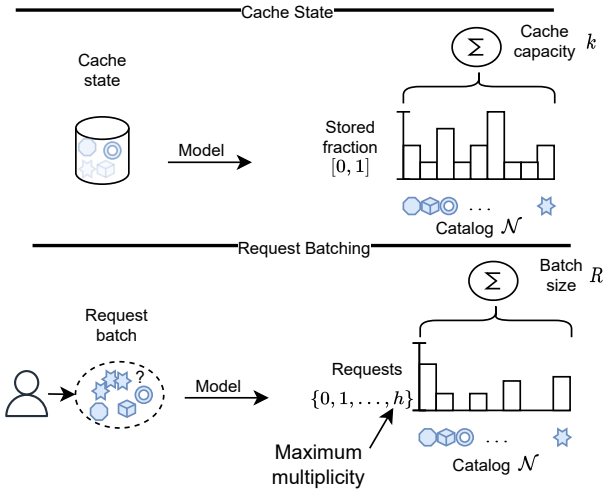
# CACHING



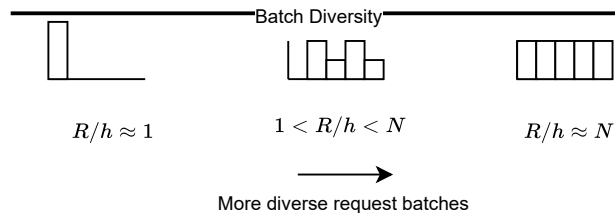
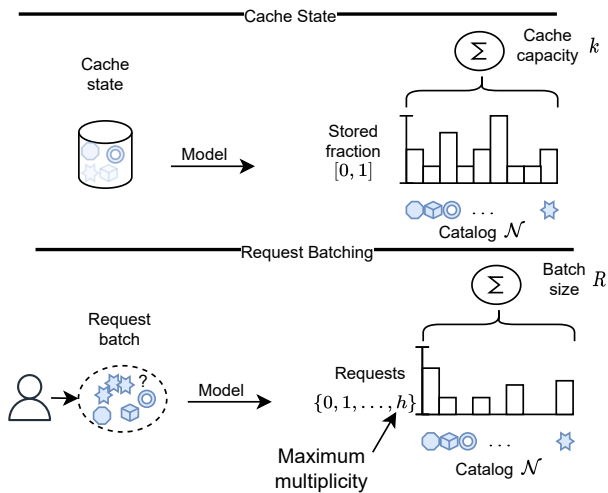
# CACHING – MODEL



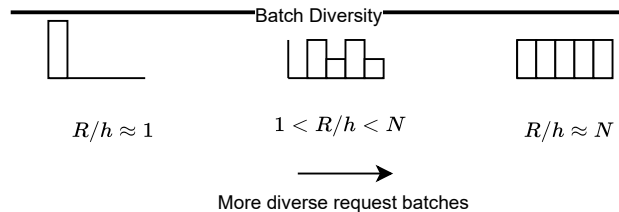
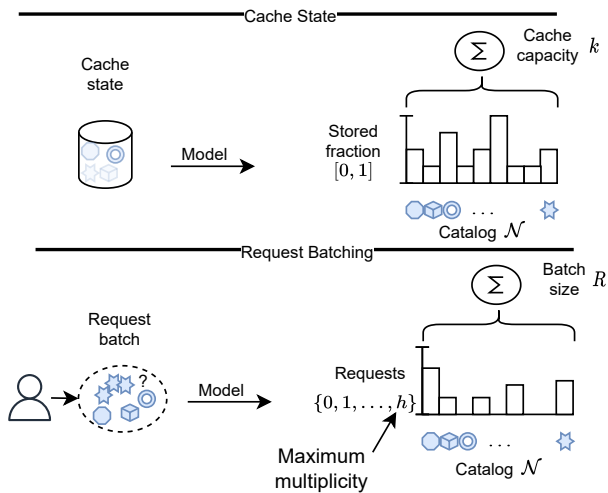
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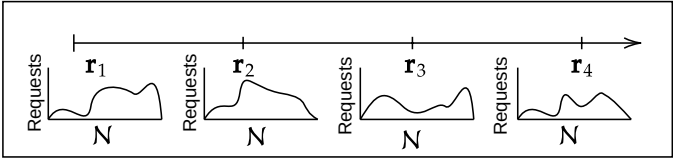
# CACHING – MODEL



When a request batch  $\mathbf{r}_t$  arrives, the cache incurs the following cost:

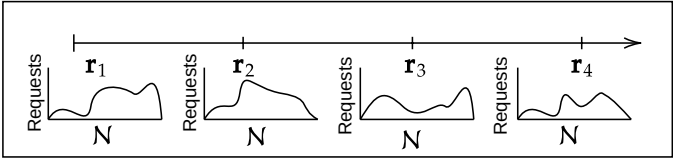
$$f_{\mathbf{r}_t}(\mathbf{x}_t) = \sum_{i=1}^N w_i r_{t,i} (1 - x_{t,i}).$$

# CACHING – SETTING



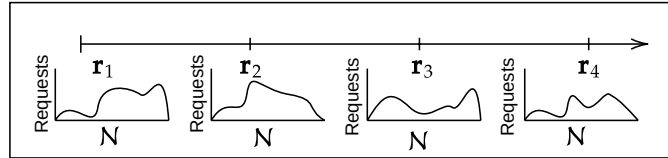


# CACHING – SETTING

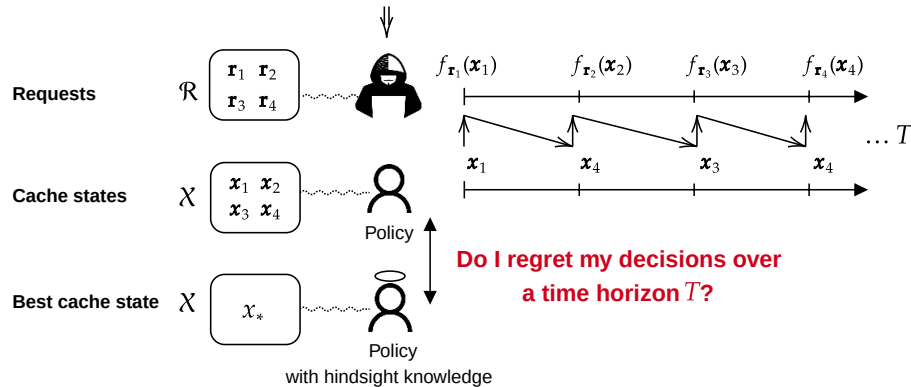


Noisy unpredictable environment can act as an **adversary** in the worst case scenario

# CACHING – SETTING



Noisy unpredictable environment can act as an **adversary** in the worst case scenario



## CACHING – PERFORMANCE METRIC

### Definition

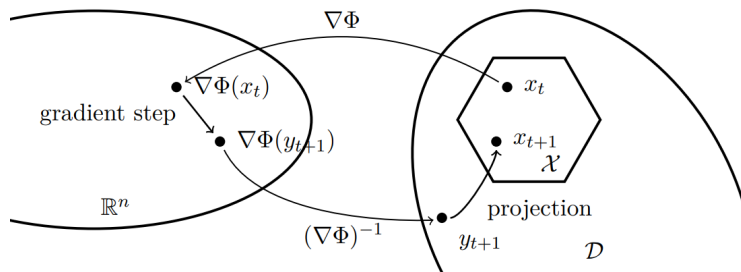
The regret of a policy  $\mathcal{A}$  is defined as

$$\text{Regret}_T(\mathcal{A}) \triangleq \sup_{\{\mathbf{r}_t\}_{t=1}^T \in \mathcal{R}^T} \left\{ \sum_{t=1}^T f_{\mathbf{r}_t}(\mathbf{x}_t) - \min_{\mathbf{x} \in \mathcal{X}} \sum_{t=1}^T f_{\mathbf{r}_t}(\mathbf{x}) \right\}.$$

When  $\text{Regret}_T(\mathcal{A})$  is sublinear in  $T$ , the policy  $\mathcal{A}$  experiences no regret on average as  $T \rightarrow \infty$ .

## CACHING – ONLINE MIRROR DESCENT (OMD)

A mirror map  $\Phi : \mathcal{D} \subset \mathbb{R}^{\mathcal{N}} \rightarrow \mathbb{R}$  defines a unique algorithm, e.g.,  $\Phi(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|_2^2$  defines OGD, and  $\Phi(\mathbf{x}) = \sum_{i \in \mathcal{N}} x_i \log(x_i)$  (negative entropy) defines  $\text{OMD}_{\text{NE}}$ .



**Figure.** OMD update rule (Bub15).

## CACHING – CONTRIBUTIONS

- We provide the first results to guide the selection of the best policy.

### Theorem

OGD is optimal for  $\frac{R}{h} \leq k$  (low diversity and large cache sizes). OMD<sub>NE</sub> is optimal for  $\frac{R}{h} > 2\sqrt{Nk}$  (high diversity and small cache sizes).

## CACHING – CONTRIBUTIONS

- We provide the first results to guide the selection of the best policy.

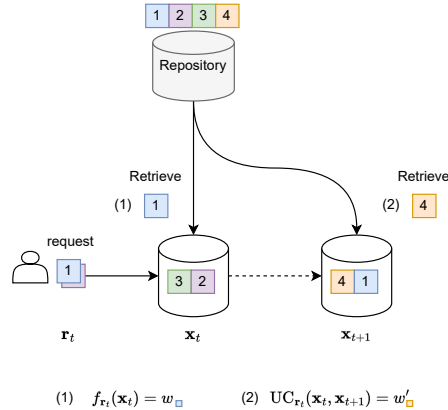
### Theorem

OGD is optimal for  $\frac{R}{h} \leq k$  (low diversity and large cache sizes).  $\text{OMD}_{\text{NE}}$  is optimal for  $\frac{R}{h} > 2\sqrt{Nk}$  (high diversity and small cache sizes).

- Highly efficient projection algorithm for  $\text{OMD}_{\text{NE}}$  that yields a policy that has the lowest time-complexity per iteration among recent works (PDVI19; PS21; BBS20; MS21).

## CACHING – UPDATE COSTS

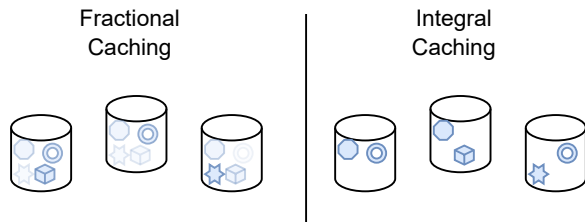
We define the update cost at time  $t$  as  $UC_{r_t}(\mathbf{x}_t, \mathbf{x}_{t+1}) \triangleq \sum_{i \notin \text{supp}(r_t)} w'_i \max\{0, x_{t+1,i} - x_{t,i}\}$ .



## Fractional Caching Incurs no Update Costs

We prove that any request batch  $r_t$ , for  $\text{OMD}_{\text{NE}}$  or  $\text{OGD}$ , it holds  $UC_{r_t}(\mathbf{x}_t, \mathbf{x}_{t+1}) = 0$ .

# INTEGRAL CACHING – NECESSITY OF RANDOMIZATION



## Proposition

Any deterministic policy restricted to select integral cache states in  $\mathcal{Z} \triangleq \mathcal{X} \cap \{0, 1\}^N$  has linear regret, i.e.,

$$\text{Regret}_T(\mathcal{A}) \geq k(1 - k/N) T.$$



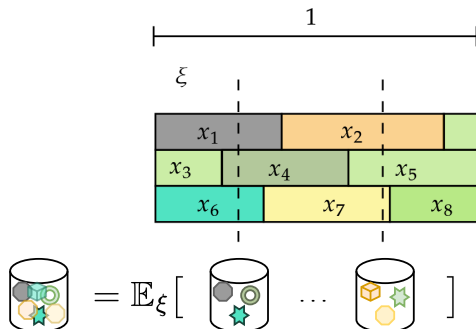
# RANDOMIZED INTEGRAL CACHING

We restrict ourselves to randomized rounding schemes that output  $z_t \in \mathcal{Z}$  such that  $\mathbb{E}[z_t] = x_t$  through some rounding  $\Xi$ .

## Remark

The expected regret  $\text{Regret}_T(\mathcal{A}, \Xi)$  is the same as the regret of  $\mathcal{A}$ .

A scheme that has this property is Madow's sampling (MM44):



# RANDOMIZED INTEGRAL CACHING

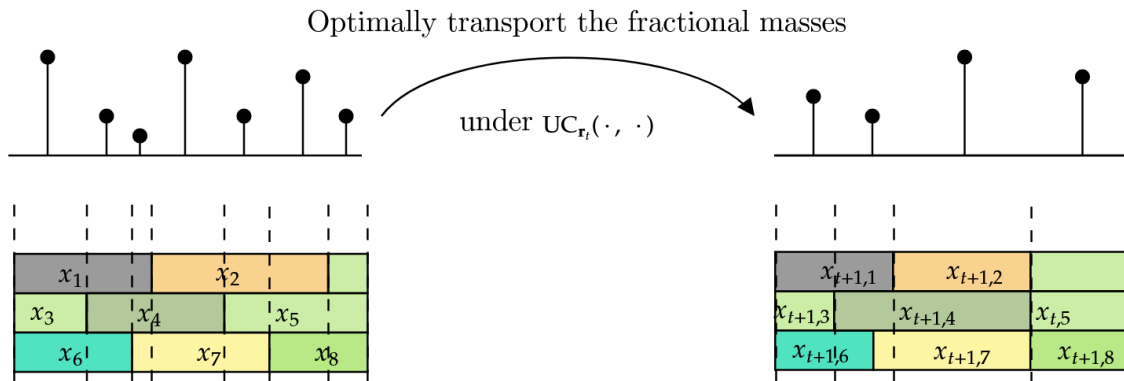
When considering the extended regret ( $\text{E-Regret}_T(\mathcal{A}, \Xi)$ ) we lose immediately the regret guarantee:

## Theorem

Any randomized caching policy constructed by an online policy  $\mathcal{A}$  combined with online independent rounding as  $\Xi$  leads to  $\Omega(T)$   $\text{E-Regret}(\mathcal{A}, \Xi)$ .

Imposing dependence (coupling) between the two consecutive random states may significantly reduce the expected update cost.

# RANDOMIZED INTEGRAL CACHING – OPTIMAL TRANSPORT

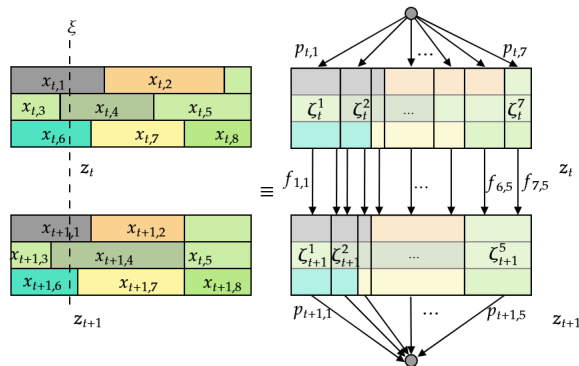


## Remark

We prove that this scheme selected as  $\Xi$ , coupled with a no-regret policy  $\mathcal{A}$ , has sublinear extended regret guarantee. **However, it has a time-complexity  $\mathcal{O}(N^3)$ .**

# RANDOMIZED INTEGRAL CACHING – SIMPLER APPROACH

Online Coupled Rounding

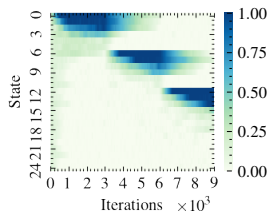


## Theorem

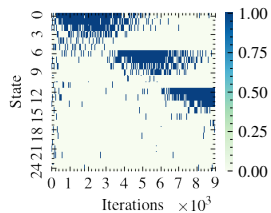
A no-regret policy  $\mathcal{A}$  combined with online coupled rounding  $\Xi$  has  $\mathcal{O}(\sqrt{T})$  E-Regret $_T(\mathcal{A}, \Xi)$ .

Online Coupled Rounding has linear time complexity.

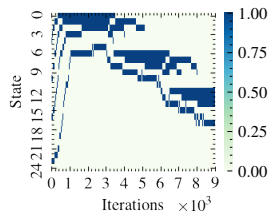
# RANDOMIZED INTEGRAL CACHING – SUMMARY



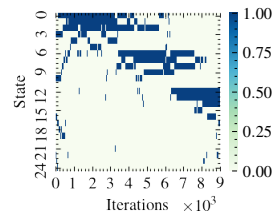
(a) Fractional cache states



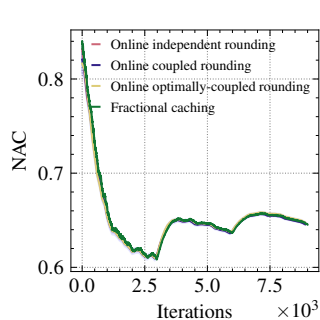
(b) Online independent rounding



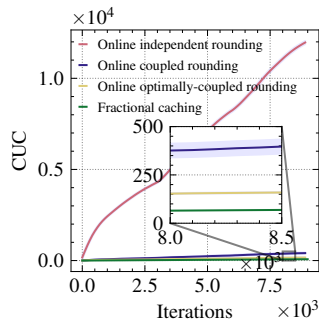
(c) Online coupled rounding



(d) Online optimally-coupled rounding



(a) Normalized average cost



(b) Cumulative update cost

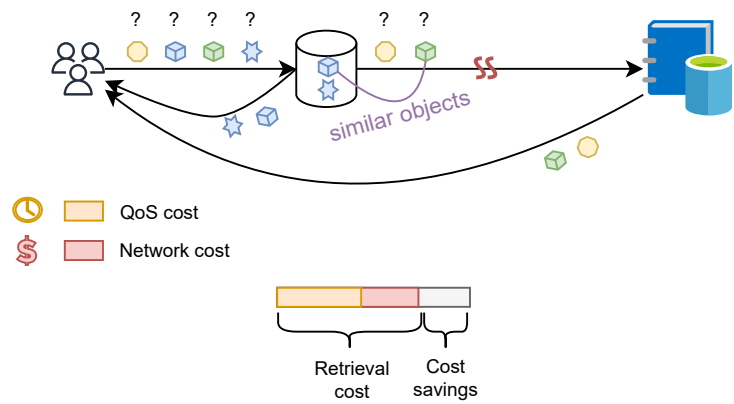
## RELATED WORK

- No-Regret Caching via Online Mirror Descent, ICC 2021 and ACM ToMPECS 2023.
- Online Caching Networks with Adversarial Guarantees, ACM SIGMETRICS, 2022.

# PRESENTATION ORGANIZATION

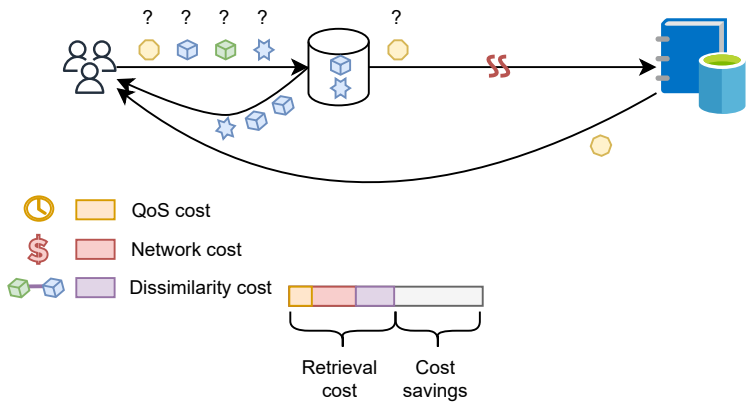
|   |                                                   |           |
|---|---------------------------------------------------|-----------|
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# SIMILARITY CACHING

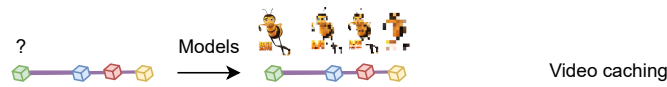




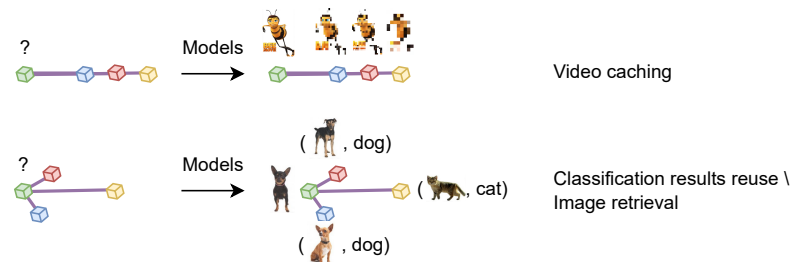
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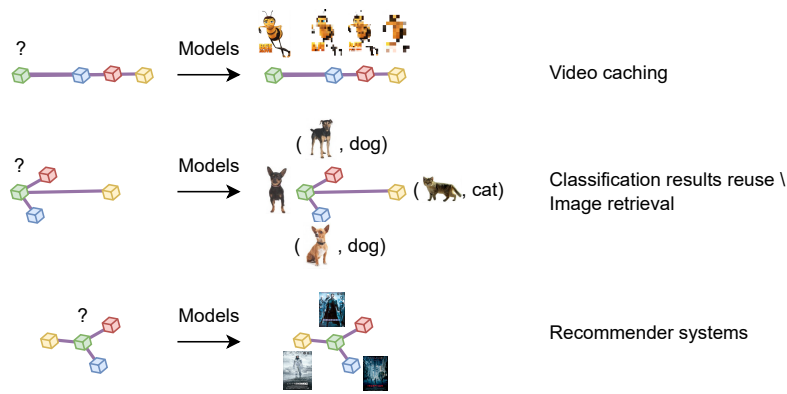
# SIMILARITY CACHING – MOTIVATION



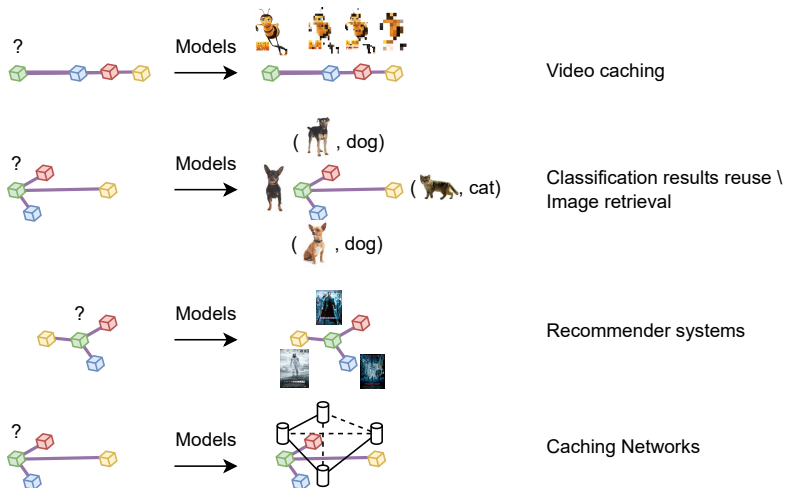
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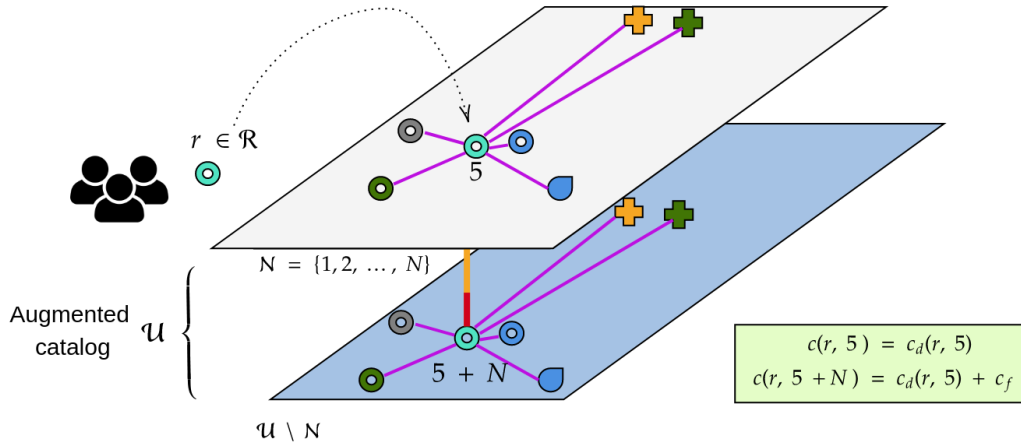


# SIMILARITY CACHING – MOTIVATION



Content recommendation (SS16; SGSV18; CS20), content retrieval (FLO<sup>+</sup>08; PBC<sup>+</sup>09), Machine Learning serving (DGT<sup>+</sup>17; DGN17; CWZ<sup>+</sup>17; VGGK18; KBVA19).

# SIMILARITY CACHING – MODEL



## SIMILARITY CACHING – CACHING GAIN

When  $r \in \mathcal{R}$  is received, a cache with allocation vector  $\mathbf{x} \in \{0, 1\}^{2N}$  incurs the cost

$$C(r, \mathbf{x}) = \sum_{i=1}^{2N} c(r, \pi_i^r) x_{\pi_i^r} \cdot \mathbb{1} \left( \sum_{j=1}^i x_{\pi_j^r} \leq k \right).$$

### Objective

Our objective is to maximize the caching gain (cost savings) as the cache state  $\mathbf{x}$  changes, given as

$$G(r, \mathbf{x}) \triangleq C(r, \text{empty cache}) - C(r, \mathbf{x}).$$

## SIMILARITY CACHING – PERFORMANCE METRIC

### Definition

The regret of the randomized policy  $\mathcal{A}$  with the cache states  $\{\mathbf{x}_t\}_{t=1}^T$  is given by

$$\psi\text{-Regret}(\mathcal{A}) = \sup_{\{r_1, r_2, \dots, r_T\} \in \mathcal{R}^T} \left\{ \max_{\mathbf{x} \in \mathcal{X}} \psi \sum_{t=1}^T G(r_t, \mathbf{x}) - \mathbb{E} \left[ \sum_{t=1}^T G(r_t, \mathbf{x}_t) \right] \right\}.$$

The constant  $\psi = 1 - 1/e$  is the best approximation ratio achievable in P to the NP-Hard static optimum

When  $\psi\text{-Regret}(\mathcal{P})$  is sublinear in  $T$ , the policy experiences *no regret* on average as  $T \rightarrow \infty$  w.r.t. the  $\psi$ -approximation of the offline problem in hindsight.



# SIMILARITY CACHING – EXPLOITING OCO

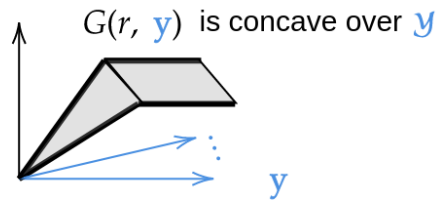
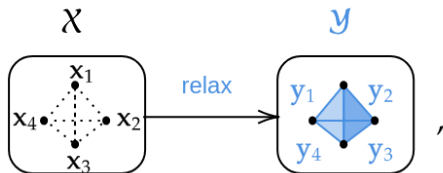
## Lemma

The caching gain can be expressed equivalently as

$$G(r, \mathbf{x}) = \sum_{i=1}^{K^r-1} \alpha_i^r \min \left\{ k, \sum_{j=1}^i x_{\pi_j^r} \right\} + G_0,$$

where  $\alpha_i^r \in \mathbb{R}_{\geq 0}$ ,  $G_0 \in \mathbb{R}$ , and  $K^r \in \mathbb{N}$  are constants.

Physical cache states    Virtual cache states



# SIMILARITY CACHING – EXPLOITING OCO

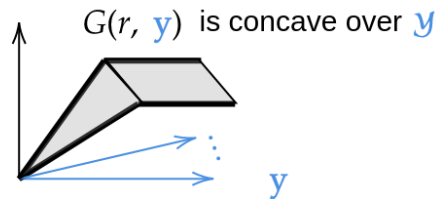
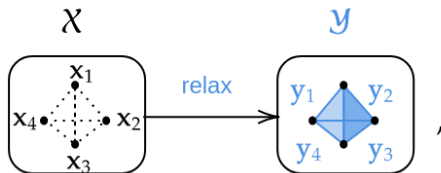
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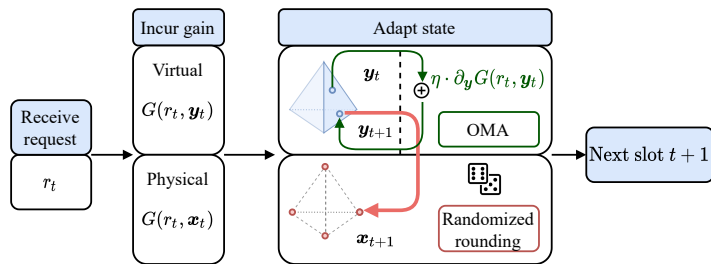
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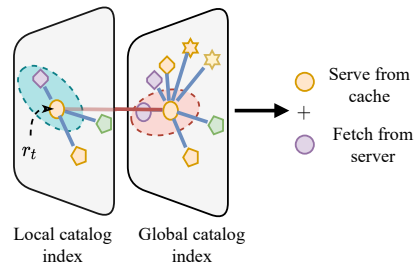
The fractionally relaxed problem can be cast in the framework of OCO (Haz16) + Exploit the property  $\mathbb{E}[G(r_t, \mathbf{x}_t)] \geq \psi G(r_t, \mathbf{y}_t)$ .

# SIMILARITY CACHING – AÇAI (ASCENT SIMILARITY CACHING WITH APPROXIMATE INDEXES) POLICY



(a)

- Dissimilarity cost  $c_d$
- Retrieval cost  $c_f$
- $k$ NN over cache index
- $k$ NN over catalog index



(b)

# SIMILARITY CACHING – AÇAI POLICY PERFORMANCE GUARANTEES

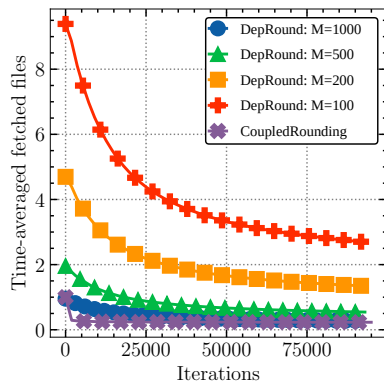
## Theorem

AÇAI configured with a negentropy mirror map, learning rate  $\eta^*$ , and rounding scheme ElasticCoupledRounding or DepRound with a freezing period  $M = \Theta(T^\beta)$  for  $\beta \in [0, 1)$  satisfies

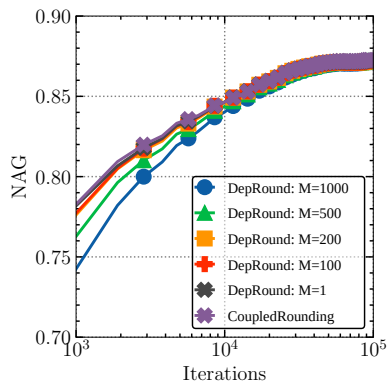
$$(1 - 1/e)\text{-Regret}_{\mathcal{X}}(\text{AAI}) = \mathcal{O}\left(T^{\frac{1+\beta}{2}}\right).$$

The parameter  $M$  reduces cache updates at the expense of reducing the cache reactivity. The update cost  $\mathcal{C}_{\text{UC},T}$  is given as  $\mathcal{C}_{\text{UC},T} = \mathcal{O}(T^{1-\beta})$  for DepRound and  $\mathcal{C}_{\text{UC},T} = \mathcal{O}(\sqrt{T})$  for ElasticCoupledRounding.

# SIMILARITY CACHING – SERVICE/UPDATE COSTS TRADEOFF

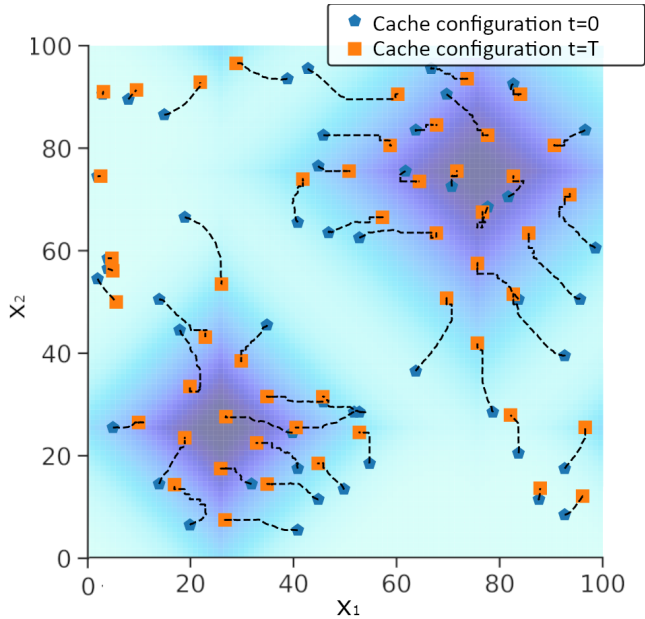


(a) Time-averaged fetched files



(b) Caching gain

# SIMILARITY CACHING – A HEURISTIC UNDER CONTINUOUS CATALOGS



## Caching Scheme

GRADES heuristic uses gradient descent to navigate the continuous space and find appropriate objects to store in the cache.

## RELATED WORK

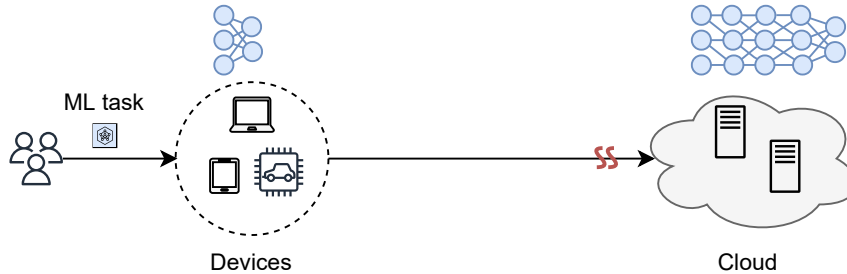
- GRADES: Gradient Descent for Similarity Caching, IEEE INFOCOM, 2021 and IEEE/ACM ToN, 2022.
- AÇAI: Ascent Similarity Caching with Approximate Indexes ITC 33, 2021 and IEEE/ACM ToN, 2022.

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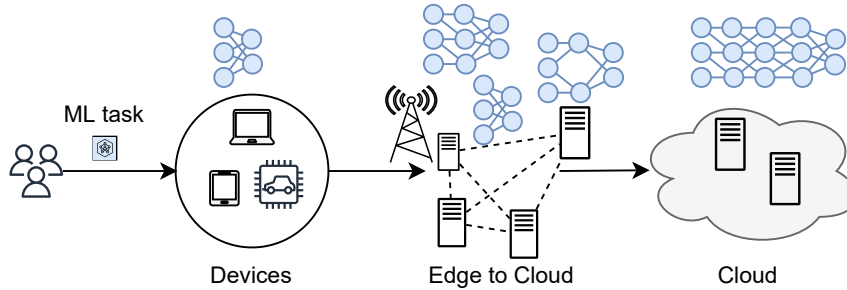
# INFERENCE DELIVERY NETWORKS



## Current ML deployment

Simpler models available locally have low accuracy. Complex models in the cloud may introduce high latency

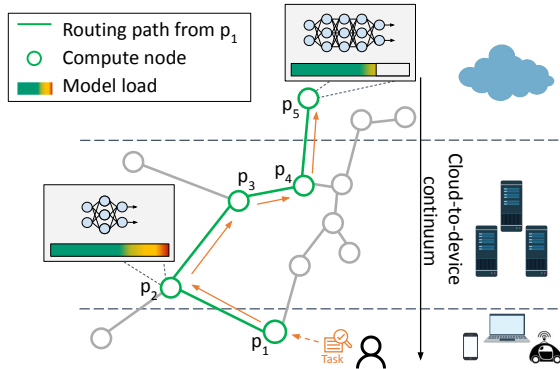
# INFERENCE DELIVERY NETWORKS



## IDNs

Integrate ML inference in the continuum between end-devices and the cloud.

# INFERENCE DELIVERY NETWORKS – MODEL



## Differences with Vanilla Similarity Caching

- Models have serving capacity, and can saturate when their capacity is exceeded
- Distribute allocation decisions among computing nodes with limited information exchange

# INFERENCE DELIVERY NETWORKS – CONTRIBUTIONS

## Contribution

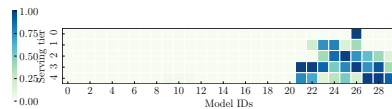
We propose a distributed online allocation algorithm for IDNs with a  $\psi$ -regret guarantee.



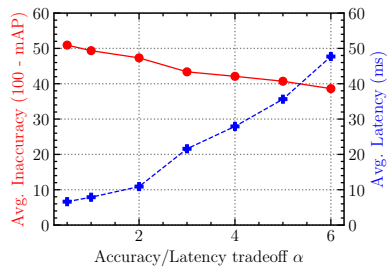
(a)  $\alpha = 3$



(b)  $\alpha = 4$



(c)  $\alpha = 5$



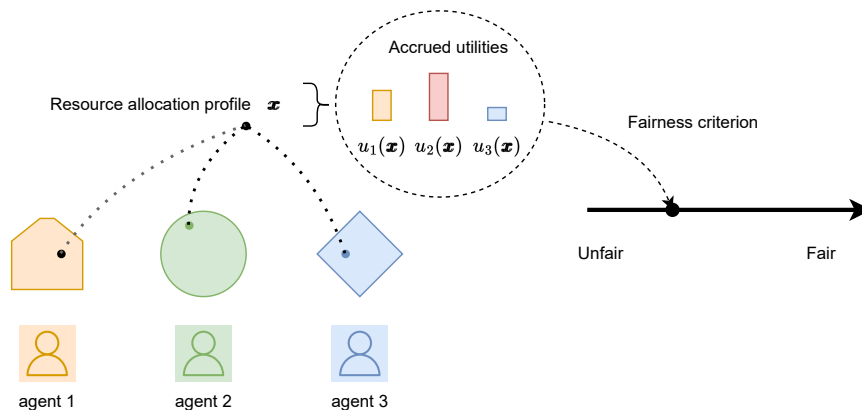
## RELATED WORK

- Towards Inference Delivery Networks: Distributing Machine Learning with Optimality Guarantees MedComNet, 2021 and IEEE/ACM ToN, 2023.

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# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION



- Consider a system comprised of a set of agents  $\mathcal{I}$ .
- An allocation profile is selected from a convex set  $\mathcal{X} = \times_{i \in \mathcal{I}} \mathcal{X}_i$ .
- The utility of each agent  $i$  is a concave function  $u_i(\mathbf{x})$ .
- An  $\alpha$ -fairness criterion is applied over the vector valued utilities  $\mathbf{u}(\mathbf{x}) = (u_i(\mathbf{x}))_{i \in \mathcal{I}}$ .

## FAIRNESS IN DYNAMIC RESOURCE ALLOCATION – APPLICATIONS

The  $\alpha$ -fairness function  $F_\alpha : \mathcal{U} \rightarrow \mathbb{R}$  is parameterized by the *inequality aversion parameter*  $\alpha \in \mathbb{R}_{\geq 0}$ , and it is given by

$$F_\alpha(\mathbf{u}) \triangleq \begin{cases} \sum_{i \in \mathcal{I}} \frac{u_i^{1-\alpha} - 1}{1-\alpha}, & \text{for } \alpha \in \mathbb{R}_{\geq 0} \setminus \{1\}, \\ \sum_{i \in \mathcal{I}} \log(u_i), & \text{for } \alpha = 1, \end{cases}$$

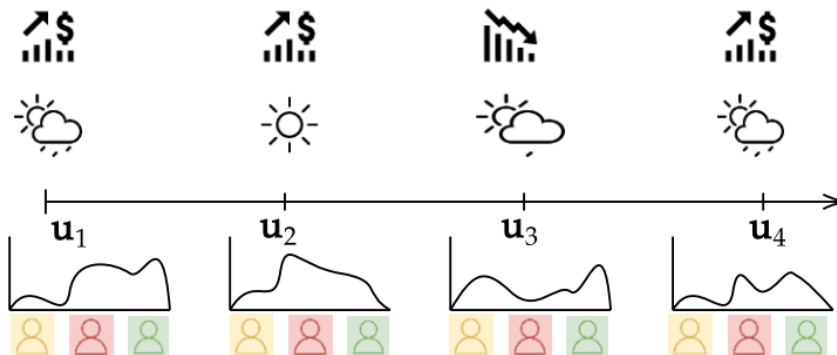
The  $\alpha$ -fairness encompasses the utilitarian principle, proportional fairness (Nash bargaining solution), max-min fairness (Kalai–Smorodinsky bargaining).

### Applications

- Congestion control in the Internet (F.P98; MW00)
- Selection transmission power in multi-user wireless networks (X. 06; Res06)
- Allocate multidimensional resources in cloud computing platforms (C. 13; W. 14; T. 15)



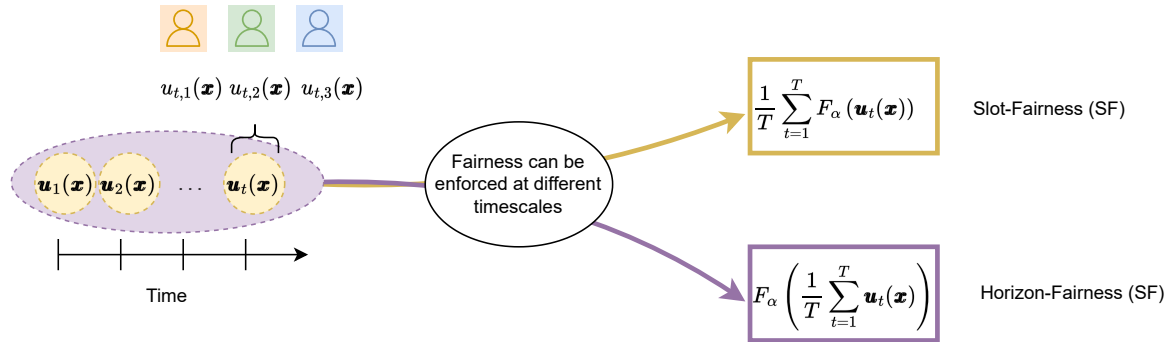
## FAIRNESS IN RESOURCE ALLOCATION — OFFLINE SETTING



Considering a *static* environment is very often unrealistic for today's communication and computing systems.

- Small-cell mobile networks
- Placing content files at edge caches
- Increasing virtualization of communication and computing systems
- Services driven by user generated data

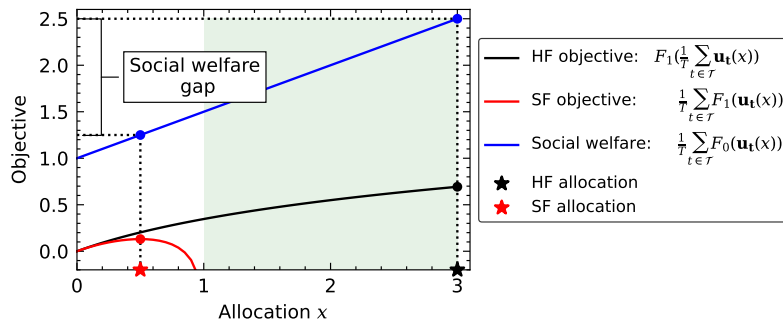
# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION – HOW TO ENFORCE FAIRNESS?



# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION

Consider a system with two agents  $\mathcal{I} = \{1, 2\}$ , an allocation set  $\mathcal{X} = [0, x_{\max}]$  with  $x_{\max} > 1$ ,  $\alpha$ -fairness criterion with  $\alpha = 1$ , even  $T \in \mathbb{N}$ , and the following sequence of utilities

$$\{\mathbf{u}_t(x)\}_{t=1}^T = \{(1+x, 1-x), (1+x, 1+x), \dots\}.$$



## Horizon-fairness/Slot-Fairness

Slot-Fairness is too restrictive.

# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION

We propose the *fairness regret* metric:

## Definition

The long-term fairness regret of a policy  $\mathcal{A}$  under  $\alpha$ -fairness is defined as follows:

$$\mathfrak{R}_T(F_\alpha, \mathcal{A}) \triangleq \sup_{\{\mathbf{u}_t\}_{t=1}^T \in \mathcal{U}^T} \left\{ F_\alpha \left( \frac{1}{T} \sum_{t \in \mathcal{T}} \mathbf{u}_t(\mathbf{x}_*) \right) - F_\alpha \left( \frac{1}{T} \sum_{t \in \mathcal{T}} \mathbf{u}_t(\mathbf{x}_t) \right) \right\}.$$

When  $\lim_{T \rightarrow \infty} \mathfrak{R}_T(F_\alpha, \mathcal{A}) \leq 0$ , policy  $\mathcal{A}$  will attain match the fairness objective of the offline benchmark under any possible sequence of utility functions.

# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION

## Impossibility Result

We prove that vanishing regret cannot be achieved in presence of an unrestricted adversary (as the one assumed in OCO).

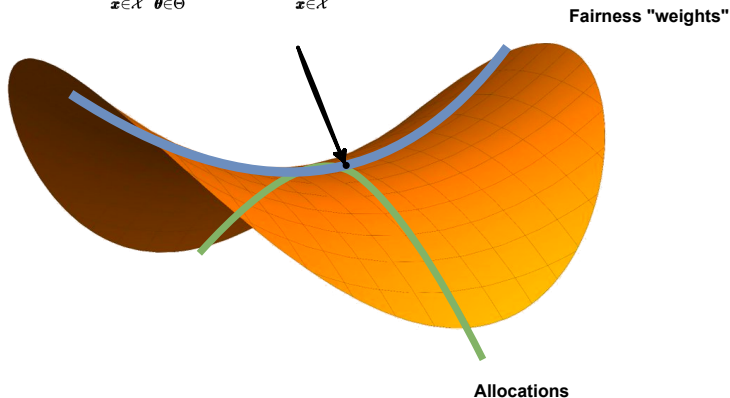
We prove that *mild* restrictions on the adversary's capabilities make vanishing regret achievable. We provide an online policy that indeed guarantees vanishing regret under these restrictions.

## Necessary Restrictions

These restrictions capture several practical utility patterns, such as non-stationary corruptions, ergodic and periodic inputs (LGK22; BLM22; ZLL<sup>+</sup>19; DAJJ12).

# ONLINE HORIZON FAIR (OHF) POLICY

$$\max_{\mathbf{x} \in \mathcal{X}} \min_{\boldsymbol{\theta} \in \Theta} \Psi(\boldsymbol{\theta}, \mathbf{x}) = \max_{\mathbf{x} \in \mathcal{X}} F_{\alpha}(\mathbf{u}(\mathbf{x}))$$



Our policy employs a *time-varying* a convex-concave function:

$$\Psi_{t,\alpha}(\boldsymbol{\theta}, \mathbf{x}) \triangleq \underbrace{(-F_{\alpha})^*}_{(I)}(\boldsymbol{\theta}) - \underbrace{\boldsymbol{\theta} \cdot \mathbf{u}_t(\mathbf{x})}_{(II)}$$

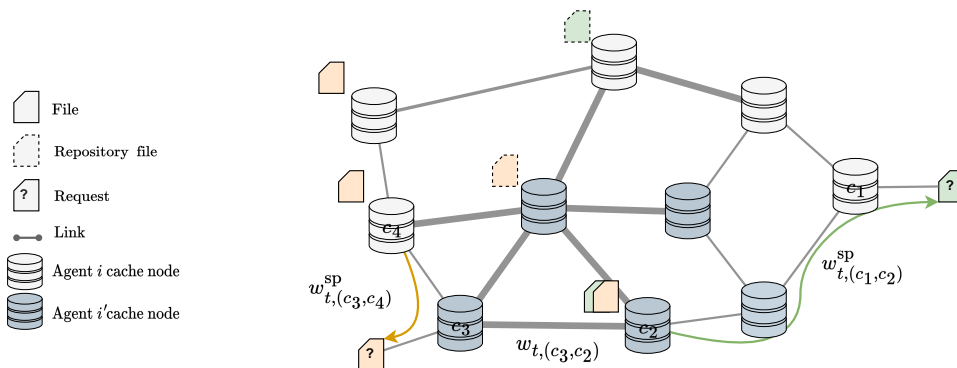
(I) A *convex conjugate* term that tracks the global fairness metric in a dual (conjugate) space

(II) A weighted sum of utilities term that tracks the appropriate allocations in the primal space

OHF policy attains the following fairness regret guarantee:

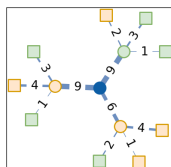
$$\Re_T(F_{\alpha}, \mathcal{A}) = \mathcal{O}\left(\frac{1}{\sqrt{T}} + \frac{\min\{\mathbb{V}_{\mathcal{T}}, \mathbb{W}_{\mathcal{T}}\}}{T}\right) = o(1).$$

# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION

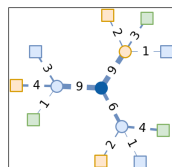


An application: a network comprised of a set of caching nodes  $\mathcal{C}$ . A request arrives at a cache node  $c \in \mathcal{C}$ , it can be partially served locally, and if needed, forwarded along the shortest retrieval path to another node to retrieve the remaining part of the file.

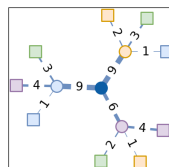
# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION – SOME RESULTS



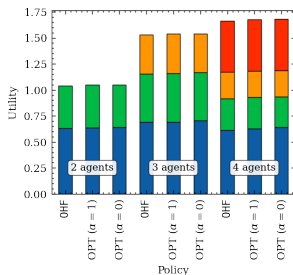
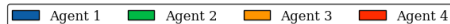
(a) TREE-1



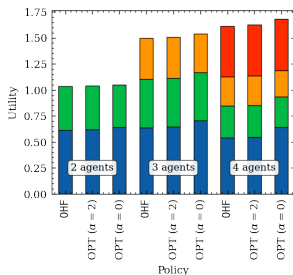
(b) TREE-2



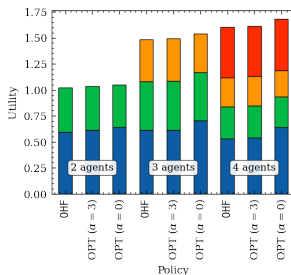
(c) TREE-3



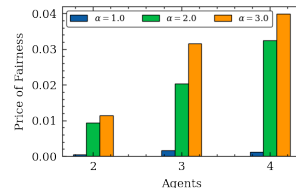
(a)  $\alpha = 1$



(b)  $\alpha = 2$



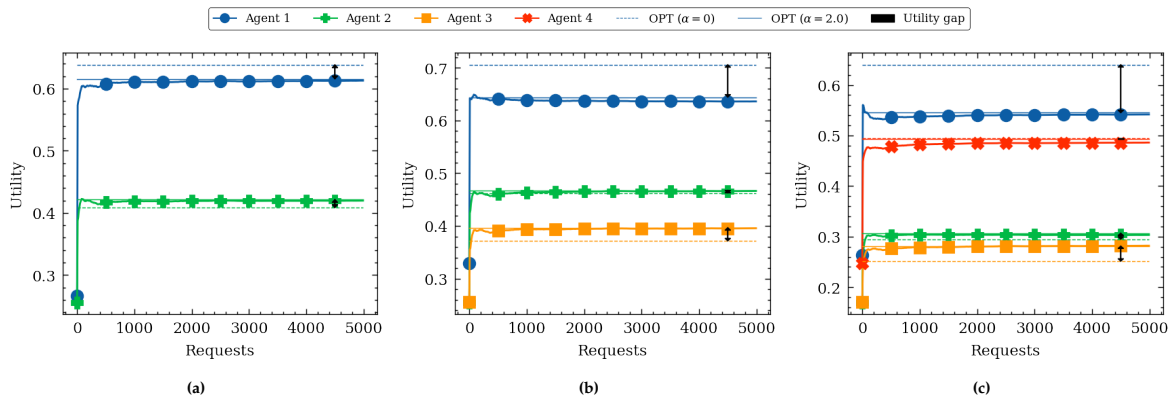
(c)  $\alpha = 3$



(d) Price of Fairness

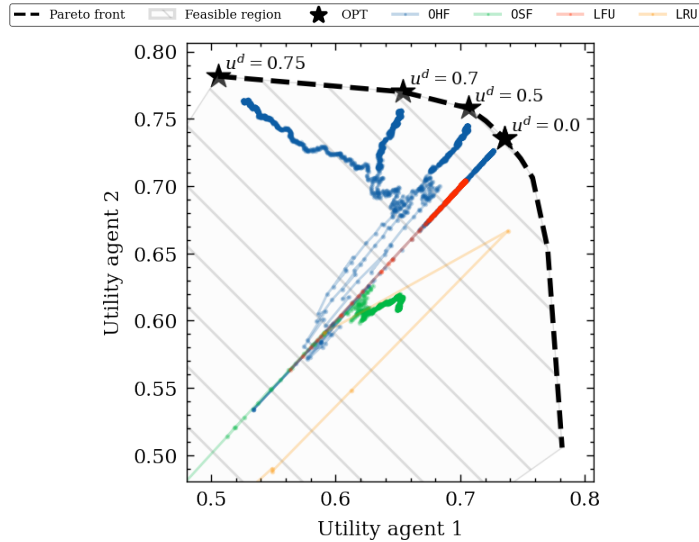


# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION – SOME RESULTS



The time-averaged utility across different agents obtained by OHF policy and OPT for  $\alpha = 2$  under an increasing number of agents in  $\{2, 3, 4\}$  and TREE 1–3 network topology.

# FAIRNESS IN DYNAMIC RESOURCE ALLOCATION – SOME RESULTS



## RELATED PAPER

- Enabling Long-term Fairness in Dynamic Resource Allocation ACM SIGMETRICS, 2023.

# PRESENTATION ORGANIZATION

- 1 Network Resource Allocation . . . . . 3
- 2 Caching . . . . . 5
- 3 Similarity Caching . . . . . 27
- 4 Inference Delivery Networks . . . . . 44
- 5 Fairness in Dynamic Resource Allocation . . . . . 50
- 6 Concluding Remarks . . . . . 64**

## CONCLUDING REMARKS

- We demonstrated the versatility of gradient algorithms on inherently combinatorial problems when paired with an opportune randomized rounding scheme.
- Our extensive experimental findings support the thesis that these algorithms are robust and can adapt to changing external system's parameters.
- We proposed a novel long-term online fairness framework for settings where the agents' utilities are subject to unknown, time-varying, and potentially adversarial perturbations.

## POTENTIAL FUTURE WORK

- Investigate dimensionality reduction techniques to diminish the operational complexity of online learning algorithms.
- Bridge the horizon-fairness and slot-fairness criteria to target applications where the agents are interested in ensuring fairness within a target time window.
- Add support for coalition formation in our fairness framework.
- Consider a limited feedback scenario where only part of the utility is revealed to the agents.

## CONCLUDING REMARKS

Thank you for your attention.

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